F17PP456-VS
VARIABLE SPEED DRIVE

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10002

RD WH BU
11,000V
FCF 160A
QAC 160A
400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10011 SHEET 14

L1 L2 L3 — LINE SIDE INCOMING SUPPLY SHEET 6

T1 T2 T3 — OUTGOING MOTOR SUPPLY SHEET 6

CABLE SCREEN EARTHED

F17-PP-456
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
3,300V
800 kW

F17-PP-456
MOTOR HEATER TERMINAL BOX
X2 54 53
F17PP456HT-L
F17PP456HT-N
0.3kW
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6

X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY SHEET 6

F17-PP-456
MOTOR RTD TERMINAL BOX

PT100 RTD INPUT CH1	+ 13	F17PP456A-RD	MOTOR WINDING RTD (RD PHASE) (TD1)
	COM 14	F17PP456A-WH	
	- 15	F17PP456A-BK	
PT100 RTD INPUT CH2	+ 16	F17PP456B-RD	MOTOR WINDING RTD (WH PHASE) (TD3)
	COM 17	F17PP456B-WH	
	- 18	F17PP456B-BK	
PT100 RTD INPUT CH3	+ 19	F17PP456C-RD	MOTOR WINDING RTD (BU PHASE) (TD5)
	COM 20	F17PP456C-WH	
	- 21	F17PP456C-BK	
PT100 RTD INPUT CH4	+ 22	F17PP456D-RD	MOTOR BEARING RTD (DRIVE END) (BD10)
	COM 23	F17PP456D-WH	
	- 24	F17PP456D-BK	
PT100 RTD INPUT CH5	+ 25	F17PP456E-RD	MOTOR BEARING RTD (NON DRIVE END) (BD20)
	COM 26	F17PP456E-WH	
	- 27	F17PP456E-BK	
PT100 RTD INPUT CH6	+ 28		
	COM 29		
	- 30		
PT100 RTD INPUT CH7	+ 31		
	COM 32		
	- 33		
PT100 RTD INPUT CH8	+ 34		
	COM 35		
	- 36		

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 — 1139 — X2 67
X3 17 — 9203 — X2 68 — K41 — K41 — MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 — 1179 — X2 63 — K41 — K41 — K41
X2 25 — 1199 — X2 64 — MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 — 1219 — X2 65
X2 24 — 1239 — X2 66 — K41 — K41 — K41 — MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 — 1259 — X2 70
X2 44 — 1279 — X2 69
A30 (PART OF): X2 15 — TRIP — X2 16 — ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R4 W4 B4 — 25A
415VAC
EARTH BAR
F16DB101-R4 F16DB101-W4 F16DB101-B4 F16DB101-BK4 → X4 1 2 3 4 PE1 — VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
BU BK
B3 — 10A
240VAC
EARTH BAR
F16DB201-B3 F16DB201-BK3 → X4 5 6 PE2 — VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-002
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
A2 (PART OF): DI-2B 9

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F17PP456VS-1409
F17PP456VS-1419
F17PP456VS-1429
F17PP456VS-1439

24VDC CONTROL SHEET 6
X2 13 14
F17PP456VS-1459
F17PP456VS-1469

F17PP456-LCS
LOCAL CONTROL STATION
X1 1 2 3 4 — LOS
X1 5 6 E — START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F17PP456-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A19

2000-F16-SCM-EL-10137_Rev1

Rougher & scavenger flotation F16-SW-011
F17-PP-456 flotation tails thick underflow pump 2 VSD feeder schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELVF16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10137.DWG 09SEP25 6:44AM HOTSEAT04